

Stereo Microphone Techniques Handout

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Stereo Microphone Techniques

1. Why do we need stereo microphone techniques?

Simply because the average listener hears with two ears!
One would record in stereo to:

- Simulate every day acoustic listening with two ears
- Give the listener a true directional impression.
- Improve the illusion that sound is coming from the artist/sound source when sound is synchronised to picture (movie, TV)

2. How does localisation work?

Having two ears helps us to determine the direction of sound! interaural time difference, interaural amplitude difference and spectral differences - all these factors play important parts for the brain when determining the direction of sound. In the following description, they are treated under separate headings, but when a person registers a sound, all three factors interact, helping to determine the direction from where it originates.

2.1. Interaural Time Difference (ITD)

Time lag is of particular importance when determining so-called impulse sounds like a click or a bang. If a sound comes in at an angle to the right of the face, the sound waves will not reach both ears at the same time. The time lag is due to the fact that the distance from the sound source to the left ear is a little longer than it is to the right ear. Therefore, the sound waves must travel a little longer before reaching the left ear which is farthest away.

The brain registers the time lag and informs us that the sound originates from a place a little to the right of the face. Time lag is of particular importance when determining so-called impulse sounds like a click or a bang. If a sound comes in at an angle to the right of the face, the sound waves will not reach both ears at the same time. The time lag is due to the fact that the distance from the sound source to the left ear is a little longer than it is to the right ear. Therefore, the sound waves must travel a little longer before reaching the left ear which is farthest away. The brain registers the time lag and informs us that the sound originates from a place a little to the right of the face.

2.2. Interaural Amplitude Difference IAD (aka. Intensity Difference)

When sounds are light treble sounds (over 1 kHz), the wavelength plays an essential role for the brain in determining the direction. These sounds all have a limited wavelength of less than 30 centimetres. When a person hears sounds of limited wavelengths, the head functions as a screen. If the sound comes from a place to the right of the face, the head will prevent the sound waves from reaching the left ear. Deep bass sounds, on the other hand, have a larger wavelength, and the head will not prevent the sound waves from reaching both ears.

2.3. Spectral Differences, caused by the separation effect of the head

If a sound does not come from the sides, but rather from above, below or immediately in front of the face, there is no time lag between the ears. In situations such as this, the outer ear is important as it helps determine the tone of the sound. Experience has taught us that the tone can help determine the source of the sound. People riding a motorbike wearing a helmet, for example, often find it difficult to hear where an ambulance is coming from, as the helmet reduces the outer ear's ability to determine the tone of the sound.

3. Stereo Techniques:

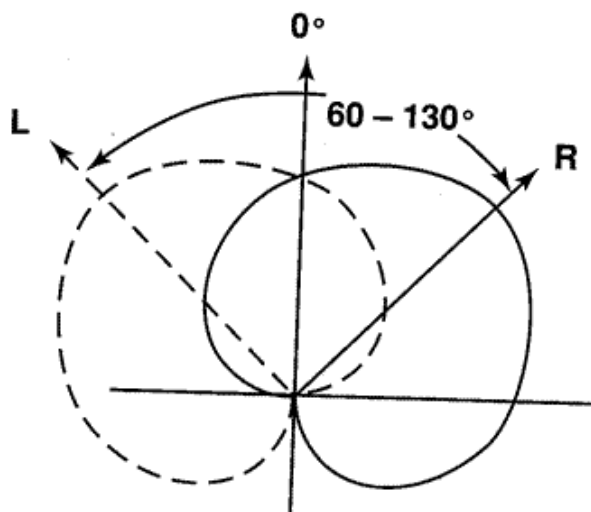
There are several stereo microphone techniques, most of them with two identical microphones. The differences between the techniques are:

- a) the polar pattern
- b) the distance between the diaphragms
- c) the angle between the microphone and the centre of the sound source
- d) the angle between the microphones
- e) the distance to the sound source

3.1. Coincident Techniques / Intensity Differences

3.1.1 XY

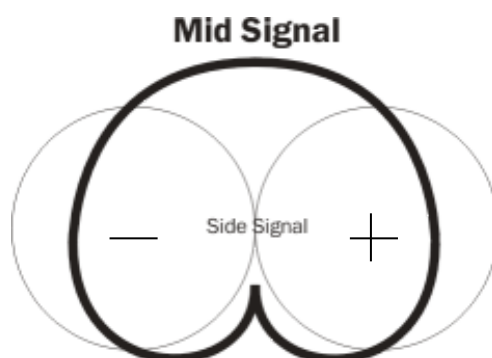
Place 2 cardioid-microphone's diaphragms as close as possible together. Depending on the cardioid type, angles between 60° and 130° are used between the microphones.



3.1.2 MS (mid – side)

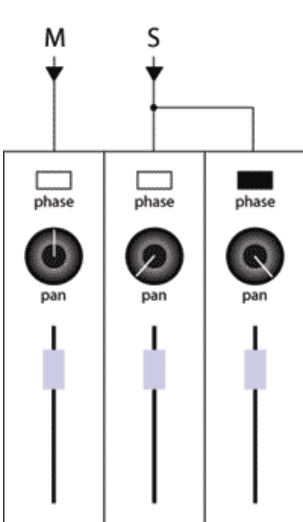
The only stereo technique with two different polar patterns. The side-microphone has to be a bi-directional (figure of eight) microphone, the mid-microphone is variable! Cardioid and omni-directional microphones are often used.

Both diaphragms are placed at the same point. The mid-microphone is pointed directly to the centre of the sound source, the side-microphone with a 90° angle.



3.1.3 MS Matrix

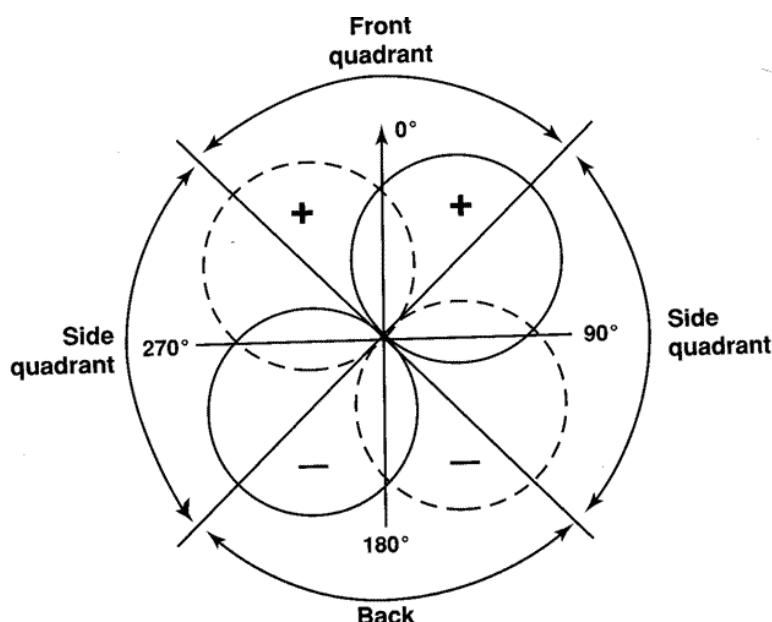
The MS signal cannot be monitored directly on a left-right system. A MS-matrix is necessary to decode the MS signal into LR signal.

	MS Matrix: The M signal is fed to one channel panned centrally. The S signal is paralleled to two channels. One channel is panned left so that it adds to the M signal. The other channel is inverted in phase and panned right so that it subtracts from the M signal. The levels of the two S signal faders should always be the same, but you can adjust both together to control the width of the stereo image. The typical level for the side signals is -3dB softer than the mid signal. But you are flexible, other levels cause different stereo-width. If you push the faders for the S channels higher than that of the M channel, you will get 'super wide' stereo. It may sound too easy to be true, but this simple stereo recording system really does work. Try it!
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Due to the presence of the centre microphone, this technique is well suited for stereo recordings where a good mono-compatibility is needed, and is extremely popular in broadcasting.

3.1.4 Blumlein

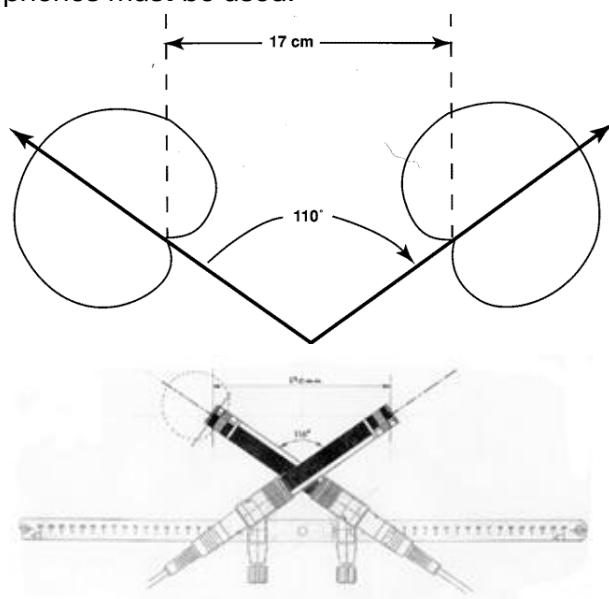
The Blumlein stereo set-up is a coincidence stereo technique, which uses two bi-directional (figure of eight) microphones in the same point and angled at 90° to each other. The Blumlein stereo is purely producing intensity related stereo information. It has the disadvantage that sound sources located behind the microphones also will be picked up and even be reproduced with inverted phase and a swapped stereo image.



3.2. Semi-Coincident Techniques

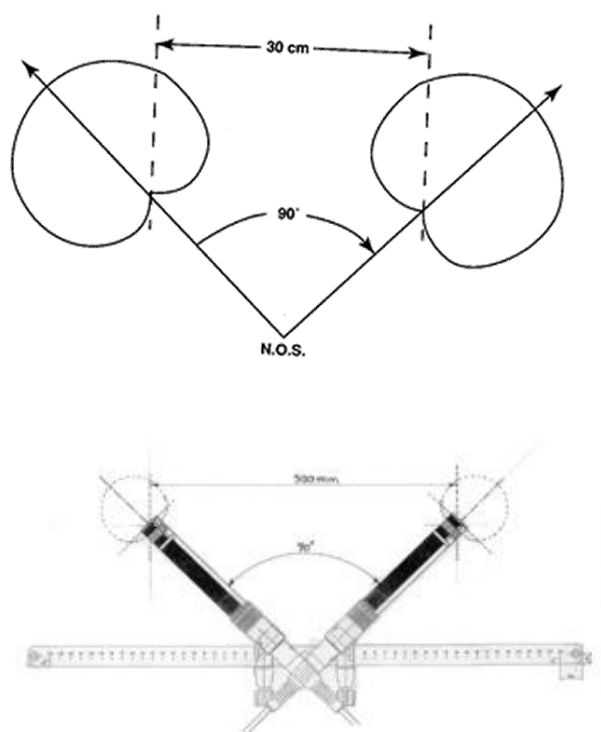
3.2.1 ORTF (Office de Radiodiffusion Télévision Française)

The default setting angle and spacing is 110° between the microphones (55° to the centre of the sound source) and 17cm between the caps. For ORTF, cardioid microphones must be used.



3.2.2 NOS (Nederlandse Omroep Stichting)

The default setting angle and spacing is 90° between the microphones (45° to the centre of the sound source) and 30cm between the caps. For NOS cardioid microphones must be used.



3.2.3 Faulkner

Faulkner is sometimes called Faulkner AB, which might be confusing: AB is a name of a spaced technique (see: 3.3.1), but Faulkner belongs to the group of semi-coincident techniques. The distance between the diaphragms is 30 cm, the microphones are arranged parallel, both microphones facing the sound source directly. The polar pattern is bi-directional (figure of eight) with the 0° angle to the source.

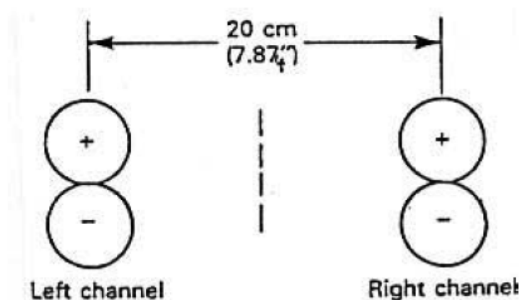


Figure 4-22 Faulkner
phased-array system

Sometimes Faulkner adds a pair of omnidirectional microphones 2 to 3 ft (60 cm to 90 cm) apart, flanking the figure eights. These outriggers add ambient spaciousness. (multi microphone techniques, see: 3.4)

3.3 Spaced Techniques

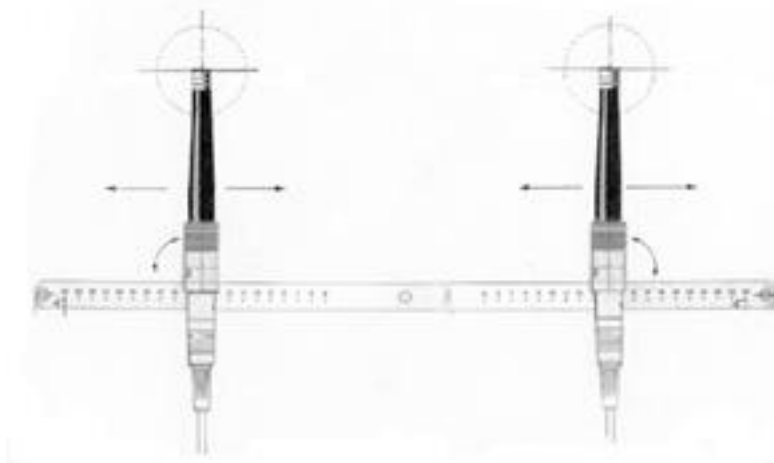
3.3.1 AB

The A-B Stereo Technique - or Time Difference Stereo, as it is sometimes called - uses two spaced (often omnidirectional) microphones to record audio signals. The microphone spacing introduces small differences in the time or phase information contained in the audio signals (according to the relative directions of the sound sources). As the human ear can sense time and phase differences in the audio signals and use them for localisation, time and phase differences will act as stereo cues to enable the listener to "capture the space" in the recording, and experience a vivid stereo image of the complete sound-field, including the positioning of each separate sound-source and the spatial boundaries of the room itself.

An important consideration when setting up for A-B stereo recordings is the distance between the two microphones. Since the acoustic character of the stereo recording is very much a question of taste, it is impossible to give fast rules for stereo microphone spacing, although it is a good idea to keep some important acoustic factors in mind.

Since the stereo width of a recording is frequency-dependent, the deeper the tonal qualities you wish to reproduce in stereo, the wider your microphone spacing should be. Using a recommended microphone spacing of a quarter of the wavelength of the deepest tone, and taking into account the human ear's reduced ability to localise frequencies below 150Hz, leads to an optimal microphone spacing of between 40 and 60 cm. Smaller microphone spacings are often used close to sound-sources to prevent the sound image of a particular musical instrument from becoming "too wide" and unnatural. Spacings down to 17 to 20 cm are detectable by the human ear, as this distance is equivalent to the distance between the two ears themselves.

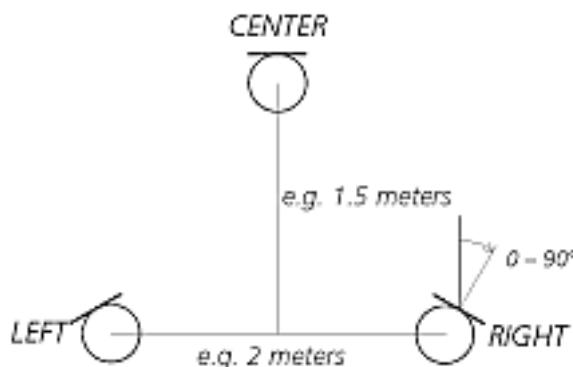
It should also be noted that an increase in microphone spacing will decrease the system's ability to reproduce the signals from sound-sources positioned directly between the microphones. This will also lead to a reduction in the quality of the stereo recording when it is played in mono.



3.4 Multiple Microphones

3.4.1 Decca Tree

The "Decca Tree" consists of a three-point pick-up by omni-directional microphones in a (often equilateral) triangle pointing at the sound source. The two outside microphones are so far apart that a "hole in the middle" will occur without adding the centre mic. The centre mic should be mixed in to fill up this hole, taking care not to make the sound perspective too mono. Outer distances are seen from 60cm to 120cm. The third, in the centre, can be slightly lower than and in front of the outside pair. Depending on the acoustic variables of the hall where the ensemble or orchestra is being recorded, the tree can be raised or lowered to achieve the best result.



3.4.2 "3 to 1" rule

The "3 to 1" rule is important for all spaced and multi-microphone techniques. The Problem:

Sound from a single source arrives at each microphone at a different time. So, a mic that is distant from the source is picking up the source with a delay, which causes variable phase shift vs. frequency. When you combine the close and distant mic signals in your mixer, certain frequencies cancel out due to phase interference, creating a "comb-filter" effect.

The solution:

Separate the mics by at least 3 times the mic-to-source distance. This creates a level difference of at least 9 dB between microphones, which reduces the comb-filter dips to an inaudible 1 dB or less.

